conditioning/guidance of flow-matching models

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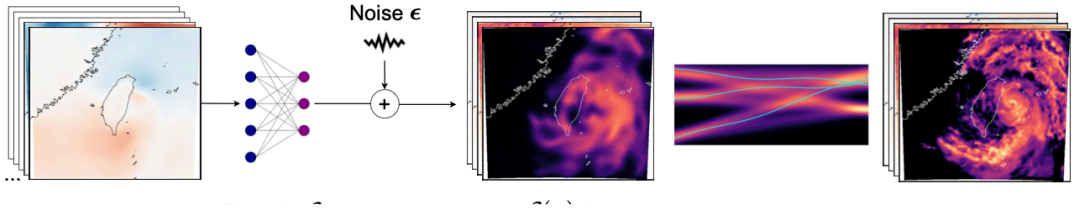
pretraining

- **adaptive flow matchencoder/embedding model**:
  - take the input (or something very simliar) through an encoder, and take the output that is closer matchted to the target as the (extra) parameter for the prediction tasks
  - great paper that did this for physical tasks:adaptive flow matching for resolving smmall-scale physics
  - opinion: useful, as it helps transform the input image to something closer of the output image, simplifying the flow matching task, **but** involving training a new model
  - 

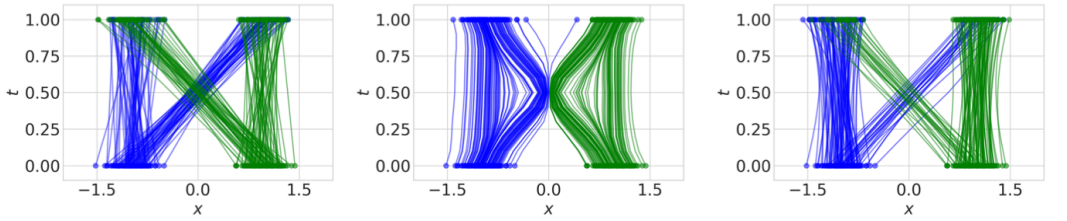
- **flowllm:** llm predicts something first, then the output is improved with guidance models
  - FlowLLM Paper
  - opinin: older paper (2024), nice approach but llm's are notoriously wonky, which can be a positive thing though. would intutitively prefer embeddings but what do i know

during training

conditioning/input data

- **conditional flow matching**
  - basically, an extra parameter, can be literally anything, usually an input image
  - opinion: bare basic conditioning, nothing to see here, basically just input -> output mapping

algorithms/optimizing function

- **variational rectified flow matching**: basically making paths more "natural" and less deterministic
  - paper by apple: Variational Rectified Flow Matching
  - opinion: might be useful to improve results, as biological processes are not deterministic/straight path
  - 

(a) Ground Truth (b) Rectified FM (Baseline) (c) Variational Rectified FM (Ours)
- **physics informed guidance**:famework designed specifically to embed physical constraints, growth factors, differential equations etc
  - fine-tune based on pdes, for example: https://www.arxiv.org/abs/2508.09156
  - opinion: too mathy?

multiple images of in-vitro dataset

- **deep supervision/multi marginal**: supervise intermediate steps as well, ensuring faster convergence, according to bytedance
  - deepflow paper
  - opinion: bytedance has a high standard, might be interesting, fits the problem well when we have multiple images
  - maybe related approach, multi marginal they call it here,
  - another paper calling the approach multi marginal
  - deep supervision and multi marginal might bet two different things
- **trajectory matching**: modelling (geographical) trajectories via flow-matching, basically predicting a path
  - two papers, 1 and 2 that do this
  - opinion: could be interesting if we think of single neurons as path, but dont know how it scales with neuronal morphologies
- **video generation** modelling the problem as a video problem might be useful when we have multiple images of the neuroink growth

reinforcement learning

- reinforcement learning improved adherence to prompt/images/text guidance, "Each denoising step (t=10 → t=9 → t=8...) is an "action"'
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  - opinion: rl is very efficient and useful for fine-tuning, worth a look, grpo fine tuning seems very advanced

post-training

probabilistic

- **controlled variational flow matching (vfm)**: including bayesian inference and making 'uncontrolled models' controlled post-training:
  - controlled generaation with equivariant variational flow matching paper, icml 2025
  - opinion: useful as an alternative to controlnet? taking a big trained model that works and controlling it's predictions
    - opinion of te authors as well: `By leveraging pretrained, unconditional generative models as priors and applying inference-based techniques like MCMC or energy-based approaches for task-specific fine-tuning, we can create adaptable frameworks for diverse applications. These hybrid strategies combine the efficiency of pretrained models with the ability to incorporate detailed task-specific constraints, enabling precise control over generated outputs in complex, high-dimensional settings.`

dual-systems

- **controlnet:** train a second machine learning network, freeze the flow-matching model (flow-matching network becomes the world model, kind of) and train a new model that learns to control the model in a specific way. the controlnet takes a new input and 'controls' the flow-matching model based on that input (the new input wasnt seen by the flow-matching model during training)
  - opinion: there seems to be a lot of success and research in this area, but i

inference time

llm

- **planner-critic-approach**: planner plans -> generate an image (with a textual prompt by the planner, usually) -> generate image -> critique llm -> incorporate critique to the planners memory
  - https://arxiv.org/html/2508.11524v1
  - opinion: very dspy approach, in fact, dspy should be probably used for this, turns llms into actors/stage writers

sampling

- **diffusion elites**
  - minte carlo reward, llms, anything to reward good samples that are likely to be true
  - opinion: great test-time-inference, very adaptable, sota research
- **physics constrainted flow matching**: "PCFM enables zero-shot hard constraint enforcement for pretrained flow matching models, projecting intermediate flowstates onto constraint manifolds at inference time,"
  - opinion: basically, diffusion elites but with physics inspired sampling, if i understoof correctly